

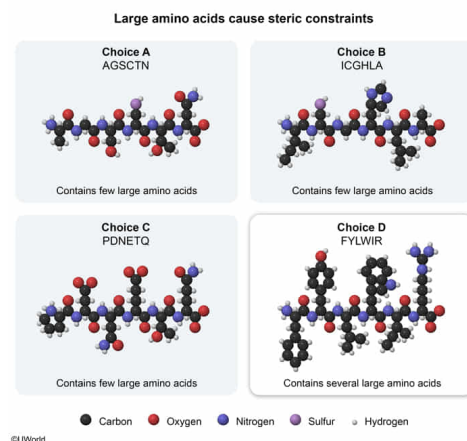
Which of the following peptides would exhibit the greatest steric constraints?

- ☐ A. AGSCTN [2%]
☐ B. ICGHLA [3%]
☐ C. PDNETQ [28%]
✓ ☒ D. FYLWIR [66%]

Correct

65% answered correctly

Explanation:



Steric effects occur within a molecule when **electron clouds** in close proximity interact and **repel each other**. Molecules (or substituent branches) that contain **many atoms** or atoms with **large electron clouds** are more **prone to experience steric effects** than molecules with few atoms or atoms that have small electron clouds. Molecular structures that are prone to steric effects are **constrained** in the **conformations they can adopt** because many conformations result in repulsive interactions and are not stable.

The **20 standard amino acids** found in natural peptides and proteins vary greatly in the sizes of their side chains, and therefore they vary in the sizes of their electron clouds. The amino acids phenylalanine (F), isoleucine (I), leucine (L), lysine (K), methionine (M), arginine (R), tryptophan (W), and tyrosine (Y) all contain **large side chains** that make substantial contributions to **steric constraints**. Peptides that contain these amino acids experience greater steric effects than peptides of the same length that contain smaller amino acids.

Each of the peptides in this question is made from six amino acids. However, the peptide **FYLWIR** consists entirely of amino acids with large side chains, and exhibits the greatest steric constraints of the choices given.

(Choice A) The amino acids in the peptide **AGSCTN** all have relatively small side chains, so this peptide does not experience much steric hinderance compared to the other choices.

(Choice B) The peptide **ICGHLA** contains some large amino acids (ie, isoleucine, leucine) but not as many as FYLWIR. Therefore, ICGHLA experiences less steric hinderance than FYLWIR.

(Choice C) The peptide **PDNETQ** contains proline, which may contribute to steric hinderance. Nevertheless, it experiences less steric hinderance than FYLWIR because, on average, the amino acids in PDNETQ have smaller side chains than those in FYLWIR.

Educational objective:

Steric effects occur when two electron clouds in close proximity repel each other. Amino acids with large side chains (eg, F, I, L, K, M, R, W, Y) contribute more to steric constraints within a peptide than other amino acids.

Time spent: 0 sec

Subject:
Biochemistry

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Amino Acids and Proteins